AutoPent

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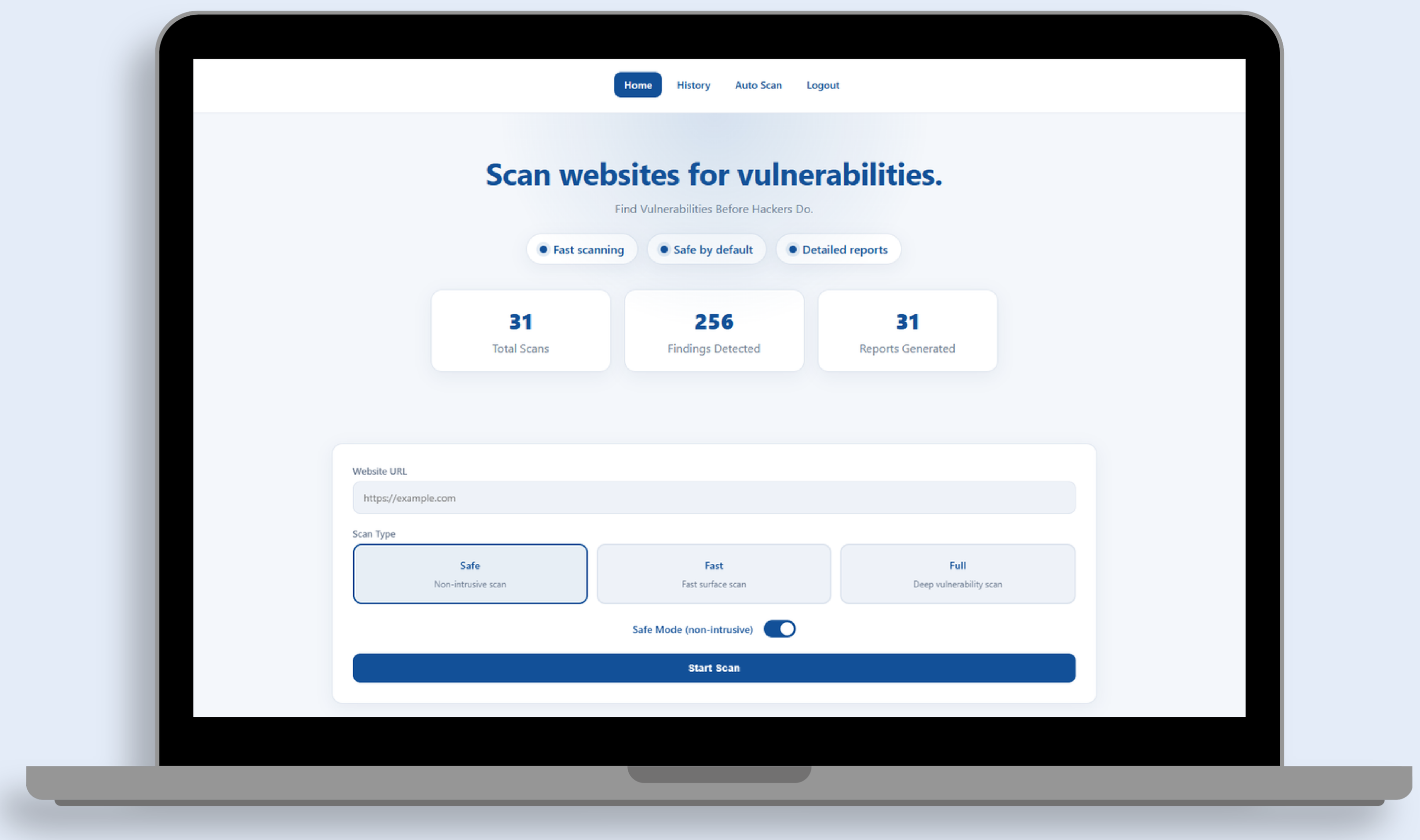
Automated Penetration Testing Tool

AutoPent is a web-based automated penetration testing tool that scans target URLs to identify potential security risks. The tool reduces manual testing effort and supports faster, more organized web security assessment.



Supervisor

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Built with

Python, GitHub, Flask, HTML, CSS, JS

Overview

As digital infrastructures grow, manual penetration testing becomes time-consuming and hard to scale. This project introduces an automated tool that detects vulnerabilities efficiently, reduces manual effort, and speeds up security assessments with clear results.

Goal

To develop an automated penetration testing tool that efficiently identifies security vulnerabilities in web applications, reduces manual effort, and provides clear, actionable results while ensuring safe and non-destructive testing.

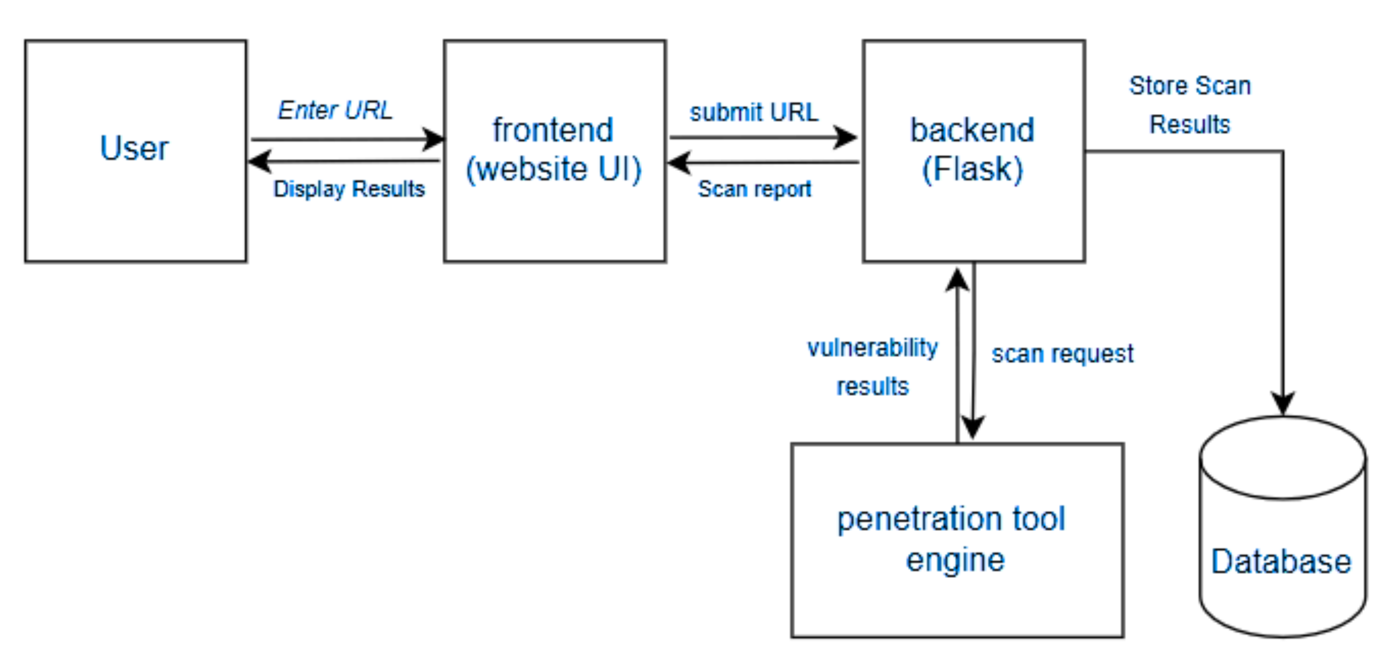
Methodology

- **Requirements Identification**: Analyzed academic studies and existing tools to identify essential features for addressing real-world cybersecurity threats.
- **System Analysis**: Defined core functionalities, including automated link verification, secure HTTP handling, and bilingual support (Arabic/English).
- **Technical Implementationn**: Developed a modular Python/Flask architecture featuring HTML-based interactive interfaces and database integration for history and monitoring.
- **Security & Evaluation:** Enforced "Safe Mode" scanning, automated Risk Scoring, results comparison, and restricted URL filtering to ensure safe and authorized testing.

Experiments

AutoPent was evaluated using multiple real-world and test websites under different scanning modes (Safe , Fast ,and Full). The experiments measured vulnerability detection, risk scoring accuracy, scan duration, and system responsiveness.The results showed that the tool effectively detected common security issues, generated clear reports, and maintained stable performance. The comparison feature also helped track changes between scans.

Project Diagram



Key Results

80%	**70%**	**High**
Faster scanning	Reduction in manual effort	Detect accuracy

Conclusion

AutoPent successfully automates web penetration testing tasks, including vulnerability detection, risk assessment, and report generation. The system enhances the usability through features such as scan history, automated scanning, and clear visualization of results.Overall, AutoPent provides an efficient and reliable solution for identifying and monitoring web security risks.